Software Requirements Specification

for

<Hospital Management System>

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of this SRS Document is to identify requirements improve functionality of Hospital Management System (HMS). From Client's viewpoint, the system aims to improve features like online appointments, managing health records. From Term Project viewpoint, it gives comprehensive and clear requirements about HMS.

1.2 Document Conventions

Abbreviation		Description
HMS	:	Hospital Management System
User	:	Patient, Doctor
SMS	:	Short Message Service
API	:	Application Programming Interface

1.3 Intended Audience and Reading Suggestions

- **Users:**
- **Intended Audience**
  These are Doctors, Patients and Admin of HMS. They interact with the functionality of HMS.
- **Reading Suggestion**
  The users focus on the interface and features of HMS.
- **Developers:**
- **Intended Audience**
  The developers focus on development and performance of HMS.
- **Reading Suggestion**
  They developers have to consider technicalities of HMS like visual diagram, coding and designing. They should also focus on both Functional and non-functional requirements.
- **Testers:**
- **Intended Audience**
  This includes Quality Assurance team that focuses on testing the features of HMS.
- **Reading Suggestions**
  They have to focus on Requirements that will help them to make Test Cases and so they can easily evaluate the performance.

- **Clients and Instructors:**
- **Intended Audience**

  This may include a consumer for which this project is being made and also supervisor who evaluates the project in educational way
- **Reading Suggestion**

  They focus on Requirements to review and confirm them.

1.4 Product Scope

- **Registration**

  This module allow user to sign up and login to HMS.
- **Appointment Management**

  This includes complete digital process of booking appointment between patients and doctors. Doctor and Patients can also view appointment schedules.
- **Patient Record**

  This module allow doctor to add and delete medical record of patients. Doctor can also view detailed list of patient's medical history.
- **Doctor Management**

  This module allows the admin of HMS to add, delete and modify profiles of doctors. The HMS includes real time dashboard to easily manage Appointment schedules and Availability of doctors.

1.5 References

The Links of Existing System Documents are as follows:

- **System # 01**

  https://github.com/alekhgupta1441/Hospital_Management_System/blob/main/Hospital%20Management%20System%20SRS.pdf
- **System # 02**

  Hospital Management System SRS for software | PDF
- **System # 03**

  https://www.scribd.com/document/624515193/Hospital-managment-system-NEWn

2. Overall Description

2.1 Product Perspective

Hospital Management System is to streamline and automate various operational tasks within a healthcare facility. It is designed to enhance the overall efficiency of hospital operations by integrating different functions and processes. The system helps in managing patient information, doctor information, scheduling appointments, maintaining medical records.

2.2 Product Functions

- **Registration**
  - Login
  - Sign up
- **Doctor Management**
  - Add Doctor Record
  - Modify Doctor List
  - View Doctor List
  - Delete Doctor Record
- **Appointment Management**
  - Book Appointment
  - Modify Appointment
  - Cancel Appointment
  - View Appointment Schedule
- **Patient Records**
  - Add Records
  - Update Records
  - View History
  - Delete Record

2.3 User Classes and Characteristics

Patient:

- **Characteristics:**
  - Patient can create account and login into system.

Doctors:

- **Characteristics:**
  - Access to patient records and view appointment schedule.

Admin:

- **Characteristics:**
  - Responsible for doctor management.

- Access to administrative functionalities such as user data updating, deletion.

2.4 Operating Environment

This Hospital Management System is compatible for both mobile and desktop devices. Our system works well with popular web-browsers like Google Chrome, Firefox and Microsoft Edge, making it easy for users to understand and interact with on the web. A stable internet connection is required to access the system features.

2.5 Design and Implementation Constraints

Technology:

The system is being developed in java net beans which highly limits the design and implementation of the system for developers and designers.

Natural Language:

The natural language of the system is English which limits the approach of the system and difficult for the people who don't speak English.

2.6 User Documentation

The user documentation provides step by step instruction on how to navigate the system, perform key-tasks, and identify common issues. It includes clear explanations of interactive examples to facilitate user-understanding.

2.7 Assumptions and Dependencies

- **Assumptions:**

- **Data Accuracy:**
  Data entered into the system by medical administrators, and other users is assumed to be accurate and reliable.
- **Security Protocols:**
  Security protocols, including user authentication and authorization measures, are in place to protect patient data and sensitive information.
- **Backup and Recovery Processes:**
  Robust backup and recovery processes are in place to safeguard data in the event of system failures or unforeseen incidents.

- **Dependencies:**
  relies on a stable and well-maintained database system to store and retrieve patient records, medical data

The success of the HMS implementation depends on effective user training programs.

3. External Interface Requirements

3.1 User Interfaces

- **Interface for Registration**

  This page contains three fields i.e. Email, username and password. Validations are also applied on these fields to ensure smooth operations and authentication.

- **Interface for Appointment**

  This page allows users to book, view and modify appointments. Patients can choose doctors according to their specialties and dashboard will display the appointment schedules of patients.

- **Interface for Patient Record**

  With this interface doctors can see medical records of patients. Doctors can access the records by entering the patient's unique id which reduces the chance of error.

- **Interface for Doctor Management**

  This Interface is only visible to admin of HMS. It allows the admin to add, delete and modify doctor's record.

3.2 Hardware Interfaces

- **Desktops/Laptops**

  This HMS is built to operate on basic laptops and desktops. It can be launched on Web Browsers like Chrome and Edge. For the smooth experience strong internet connection and sufficient RAM is required.

- **Mobile Phones**

  HMS can also be used from Mobile Phone on which its design adjusts automatically according to their screen aspect ratio. Further, Mobile Application of HMS can also be developed to enhance user experience.

3.3 Software Interfaces

- **Database**

  The HMS is connected with MySQL on which medical records, appointment schedules and login details of users are stored.

- **Operating System**

  The HMS is Compatible with all operating systems such as:

  - Windows & Linux for desktops
  - Android and iOS for mobile phones

3.4 Communications Interfaces

- **API Integration**

  HMS is also connected API's to easily communicate with other systems. HMS is connected with Payment gateways like Easypaisa, Jazzcash for secure transactions and online billing.

- **SMS Notification Systems**

  HMS facilitates the effective communication between patients and admins through notification alerts about their appointment

4. System Features

4.1 Registration

- **Functional Requirements**

  REQ-1: The system shall allow user to sign-up into the system.
  REQ-2: The system shall allow to user to log-in into the system.

4.2 Doctor management

- **Functional Requirements**

  REQ-1: The system shall allow admin to add doctor record.
  REQ-2: The system shall allow admin to modify doctor list.
  REQ-3: The system shall allow admin to view doctor list.
  REQ-4: The system shall allow admin to delete doctor record.

4.3 Appointment Management

- **Functional Requirements**

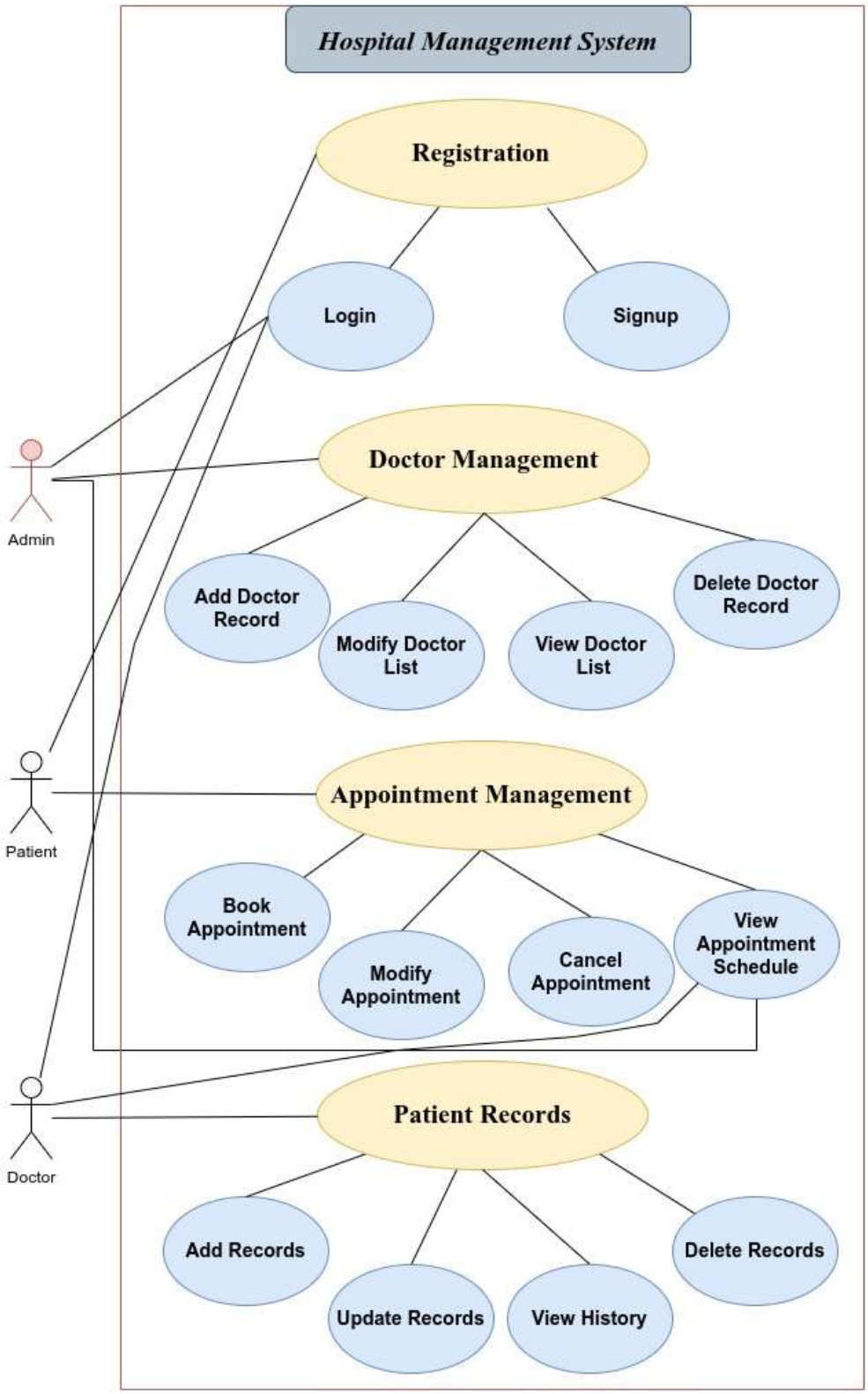
  REQ-1: The system shall allow to book appointment.
  REQ-2: The system shall allow to modify appointment.
  REQ-3: The system shall allow to cancel appointment.
  REQ-4: The system shall allow to view appointment schedule.

4.4 Patient Records

- **Functional Requirements**

  REQ-1: The system shall allow doctor to add patient records.
  REQ-2: The system shall allow doctor to update patient records.
  REQ-3: The system shall allow doctor to view record history.
  REQ-4: The system shall allow doctor to delete patient records.
  REQ-4: The system shall send remainder of appointments through SMS.

5. Use Case Diagram



6. Non-Functional Requirements

6.1 Performance Requirements

Property	Measure
Page Load Time	Pages should load in 2 seconds or less.
Concurrent Users	Should support 100+ users simultaneously.
Network Handling	Must work efficiently even on slow internet.

6.2 Safety Requirements

Property	Measure
System Stability	The system should stay stable and avoid sudden failures or crashes.
Backup Plan	Data should be regularly backed up to avoid losing important information.
Error Control	The system should manage unexpected issues without disturbing the user experience.

6.3 Security Requirements

Property	Measure
Authentication	Only authorized users can log in.
Encryption	All personal and health data should be encrypted.
Access Control	No unauthorized access or data leakage is allowed.

6.4 Software Quality Attributes

- ***Usability***

Property	Measure
User-Friendliness	Easy to use for all users (patients, doctors, admins).
Responsiveness	Interface should respond quickly.
Page Speed	Pages should load fast without delay.

- ***Reliability***

Property	Measure
Uptime	System should be online 90%+ of the time.
Stability	Should not crash or show frequent errors.
Availability	Must be accessible, especially during working hours.

- ***Maintainability***

Property	Measure
Code Simplicity	Code should be easy to understand and update.
Documentation	Clear comments and documentation should be present.
Bug Fixing	Minor bugs should be fixed in less than a day.

- ***Portability***

Property	Measure
Cross-Platform	Must work on mobile and desktop.
Browser Support	Should run on Chrome, Firefox, Edge.
Mobile App	Should be available for Android/iOS.

6.5 Business Rules

Property	Measure
Data Handling	System must follow healthcare laws .
User Roles	Users must follow specific roles (Admin, Doctor, Patient).
Login Required	Only logged-in users can access the system.